Housing Price Modeling in R: Multiple Regression, Interaction Effects, and Quadratic Regression

Project Overview

This project examines how property characteristics—including living area, number of bedrooms and bathrooms, home age, construction quality, neighborhood crime rate, school ratings, and property views—relate to residential sale prices. Multiple regression models were developed in R to estimate sale prices and evaluate the predictive importance of these characteristics. The analysis also explores interactions between qualitative variables and quadratic relationships that may not be adequately represented by a standard linear model.

Variables used in the analysis:

Variable	**What does it represent?**
price	Sale price of the home
bedrooms	Number of bedrooms
bathrooms	Number of bathrooms
sqft_living	Size of the living area in sqft
sqft_above	Size of the upper level in sqft
sqft_lot	Size of the lot in sqft
age	Age of the home
grade	Measure of craftsmanship and the quality of materials used to build the home
appliance_age	Average age of all appliances in the home
crime	Crime rate per 100,000 people
backyard	Home has a backyard (backyard=1) or not (backyard=0)
school_rating	Average rating of schools in the area
view	Home backs out to a lake (view=2), backs out to trees (view=1), or backs out to a road (view=0)

Preparing The Data Set

In [2]:
```
housing <- read.csv(file="housing_v2.csv", header=TRUE, sep=",")

# Convert appropriate variables to factors
housing <- within(housing, {
   view <- factor(view)
   backyard <- factor(backyard)
})

# number of columns
ncol(housing)

# number of rows
nrow(housing)
```

23

2692

Model #1 - First Order Regression Model with Quantitative and Qualitative Variables

First order regression model with *price* as the response variable, and *sqft_living*, *sqft_above*, *age*, *bathrooms*, and *view* as predictor variables.

In [58]:
```
# Scatterplot of Price vs. Living Area
plot(housing$sqft_living, housing$price,
     main = "Scatterplot of Price vs. Living Area",
     xlab = "Living Area", ylab = "Price",
     col="green",
     pch = 19, frame = FALSE)
```



In [59]:
```
# Scatterplot of Price vs. Age
plot(housing$age, housing$price,
     main = "Scatterplot of Price vs. Age",
     xlab = "Age", ylab = "Price",
     col="purple",
     pch = 19, frame = FALSE)
```



In [43]:
```
myvars <- c("price","sqft_living", "age")
housing_subset <- housing[myvars]

# Print the correlation matrix
print("cor")
corr_matrix <- cor(housing_subset, method = "pearson")
round(corr_matrix, 4)
```

```
[1] "cor"
```

A matrix: 3 × 3 of type dbl

	price	sqft_living	age
price	1.0000	0.6895	-0.0746
sqft_living	0.6895	1.0000	-0.3547
age	-0.0746	-0.3547	1.0000

In [3]:
```
# Create the first order regression model and print the statistics
model1 <- lm(price ~ sqft_living + sqft_above + age + bathrooms + view, data=h
ousing)
summary(model1)
```

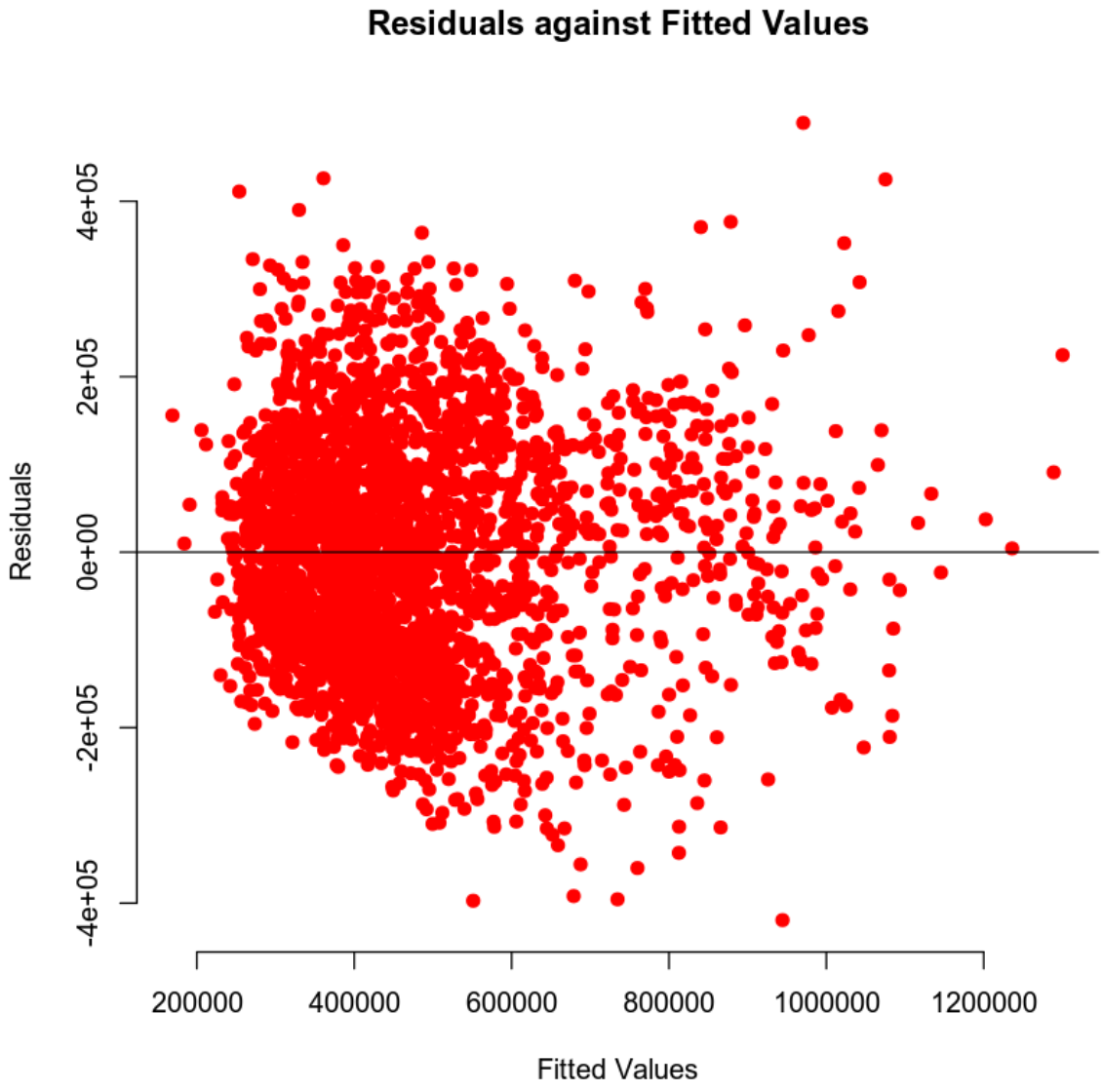
```
Call:
lm(formula = price ~ sqft_living + sqft_above + age + bathrooms +
    view, data = housing)

Residuals:
    Min      1Q  Median      3Q     Max
-419299 -101792   -5606   93896  489323

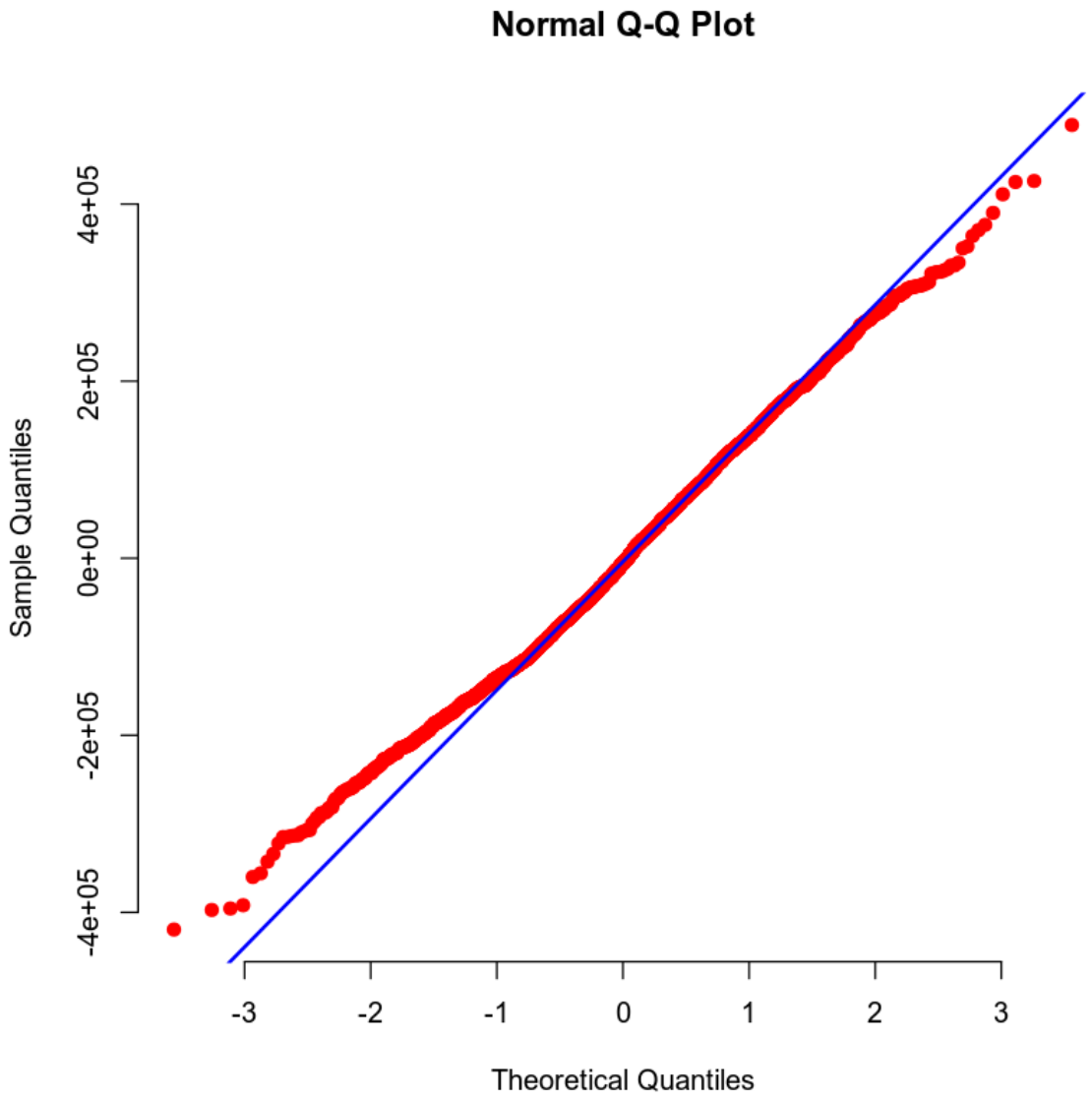
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 7.709e+03  1.411e+04   0.546  0.58495
sqft_living 1.293e+02  8.123e+00  15.916  < 2e-16 ***
sqft_above  1.951e+01  7.458e+00   2.616  0.00894 **
age         1.451e+03  1.199e+02  12.098  < 2e-16 ***
bathrooms   4.397e+04  6.126e+03   7.178 9.13e-13 ***
view1       1.675e+05  1.071e+04  15.640  < 2e-16 ***
view2       2.490e+05  1.201e+04  20.739  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 133600 on 2685 degrees of freedom
Multiple R-squared:  0.6029,    Adjusted R-squared:  0.602
F-statistic: 679.3 on 6 and 2685 DF,  p-value: < 2.2e-16
```

In [38]:
```
fitted_values <- fitted.values(model1)
residuals <- residuals(model1)
plot(fitted_values, residuals,
     main = "Residuals against Fitted Values",
     xlab = "Fitted Values", ylab = "Residuals",
     col="red",
     pch = 19, frame = FALSE)
abline(0,0)
```



In [28]:
```
qqnorm(residuals, pch = 19, col="red", frame = FALSE)
qqline(residuals, col = "blue", lwd = 2)
```



In [4]:
```
# Predicted price for a home that has 2150 sqft living area, 1050 sqft upper l
evel living area,
# is 15 years old, has 3 bathrooms, and backs out to road:
newdata <- data.frame(sqft_living=2150, sqft_above=1050, age=15, bathrooms=3,
view='0')

print("prediction interval")
prediction_pred_int <- predict(model1, newdata, interval="predict", level=0.9
0)
round(prediction_pred_int, 4)

print("confidence interval")
prediction_conf_int <- predict(model1, newdata, interval="confidence", level=
0.90)
round(prediction_conf_int, 4)
```

```
[1] "prediction interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
459828.2	239563	680093.4

```
[1] "confidence interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
459828.2	446087.9	473568.5

In [6]:
```
# predicted price for a home that has 4250 sqft living area, 2100 sqft upper l
evel living area,
# is 5 years old, has 5 bathrooms, and backs out to a lake:
newdata <- data.frame(sqft_living=4250, sqft_above=2100, age=5, bathrooms=5, v
iew='2')

print("prediction interval")
prediction_pred_int <- predict(model1, newdata, interval="predict", level=0.9
0)
round(prediction_pred_int, 4)

print("confidence interval")
prediction_conf_int <- predict(model1, newdata, interval="confidence", level=
0.90)
round(prediction_conf_int, 4)
```

```
[1] "prediction interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
1074285	852522.6	1296048

```
[1] "confidence interval"
```

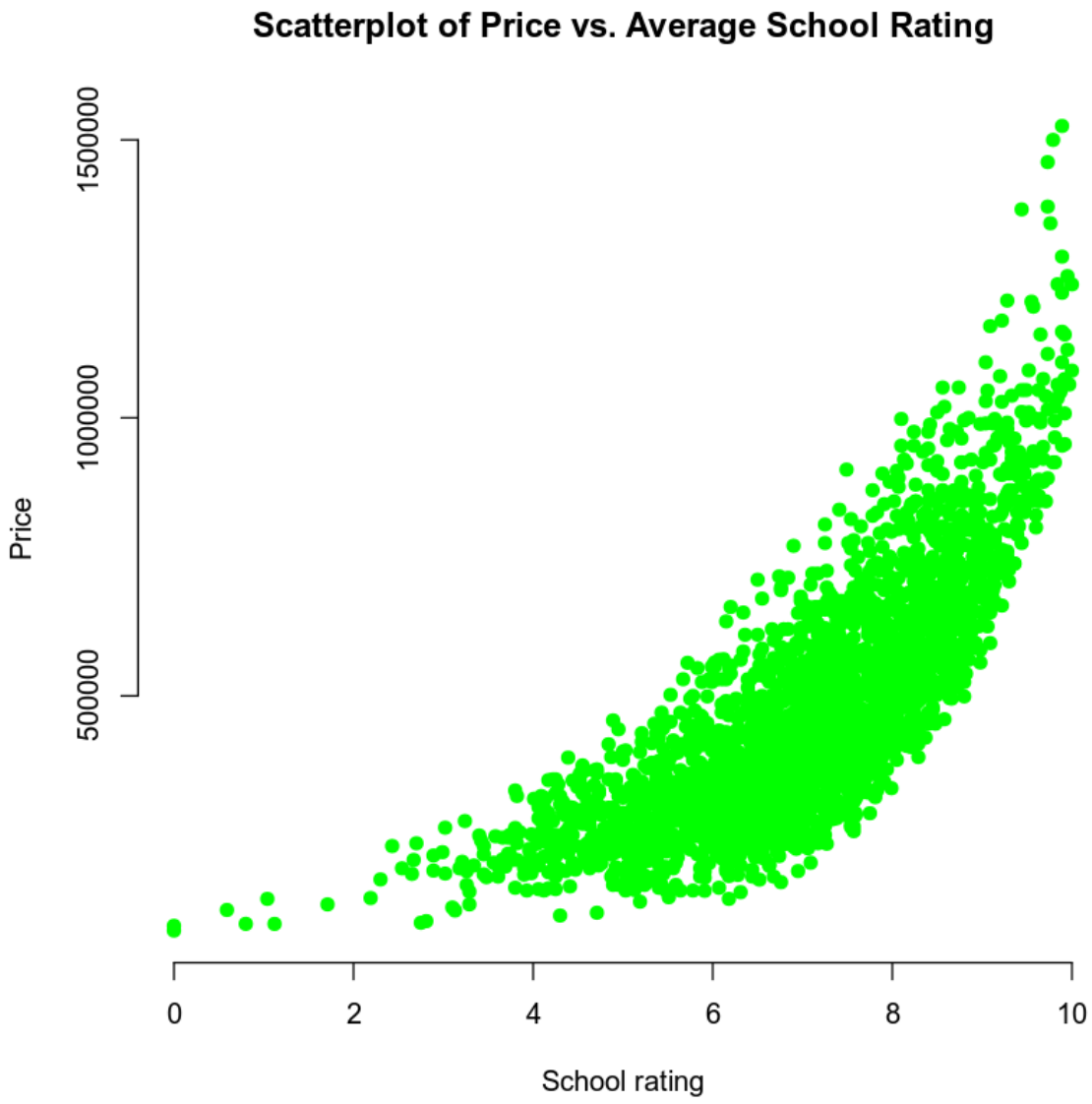
A matrix: 1 × 3 of type dbl

fit	lwr	upr
1074285	1045117	1103454

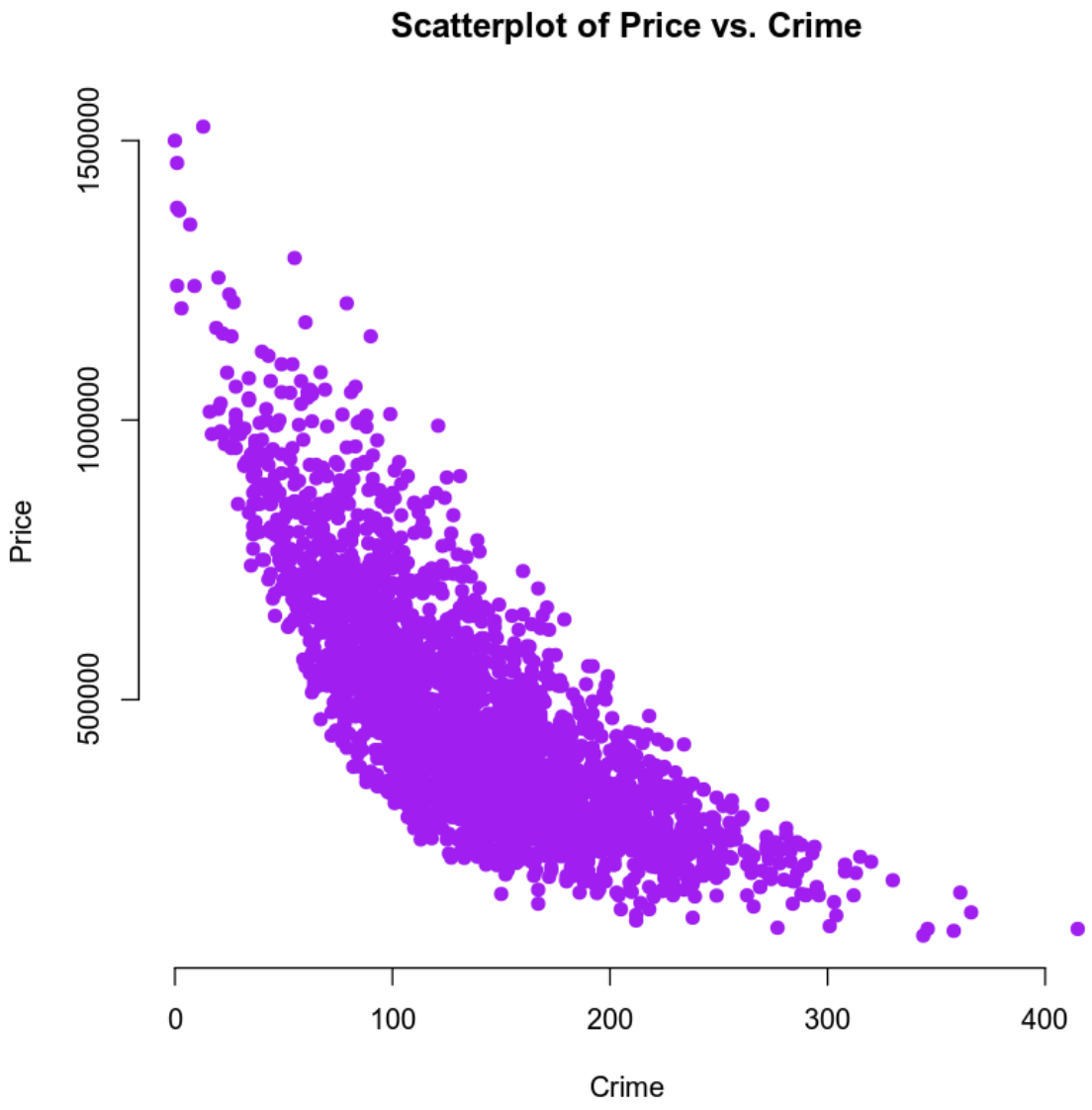
Model #2 - Complete Second Order Regression Model with Quantitative Variables

Complete second order regression model with *price* as the response variable, and *school_rating* and *crime* as predictor variables.

In [7]:
```
# Scatterplot of Price vs. Average School Rating
plot(housing$school_rating, housing$price,
     main = "Scatterplot of Price vs. Average School Rating",
     xlab = "School rating", ylab = "Price",
     col="green",
     pch = 19, frame = FALSE)
```



In [8]:
```
# Scatterplot of Price vs. Crime
plot(housing$crime, housing$price,
     main = "Scatterplot of Price vs. Crime",
     xlab = "Crime", ylab = "Price",
     col="purple",
     pch = 19, frame = FALSE)
```



In [10]:
```
# Create the second order regression model and print the statistics
model2 <- lm(price ~ school_rating + crime + school_rating:crime + I(school_ra
ting^2) + I(crime^2), data=housing)
summary(model2)
```

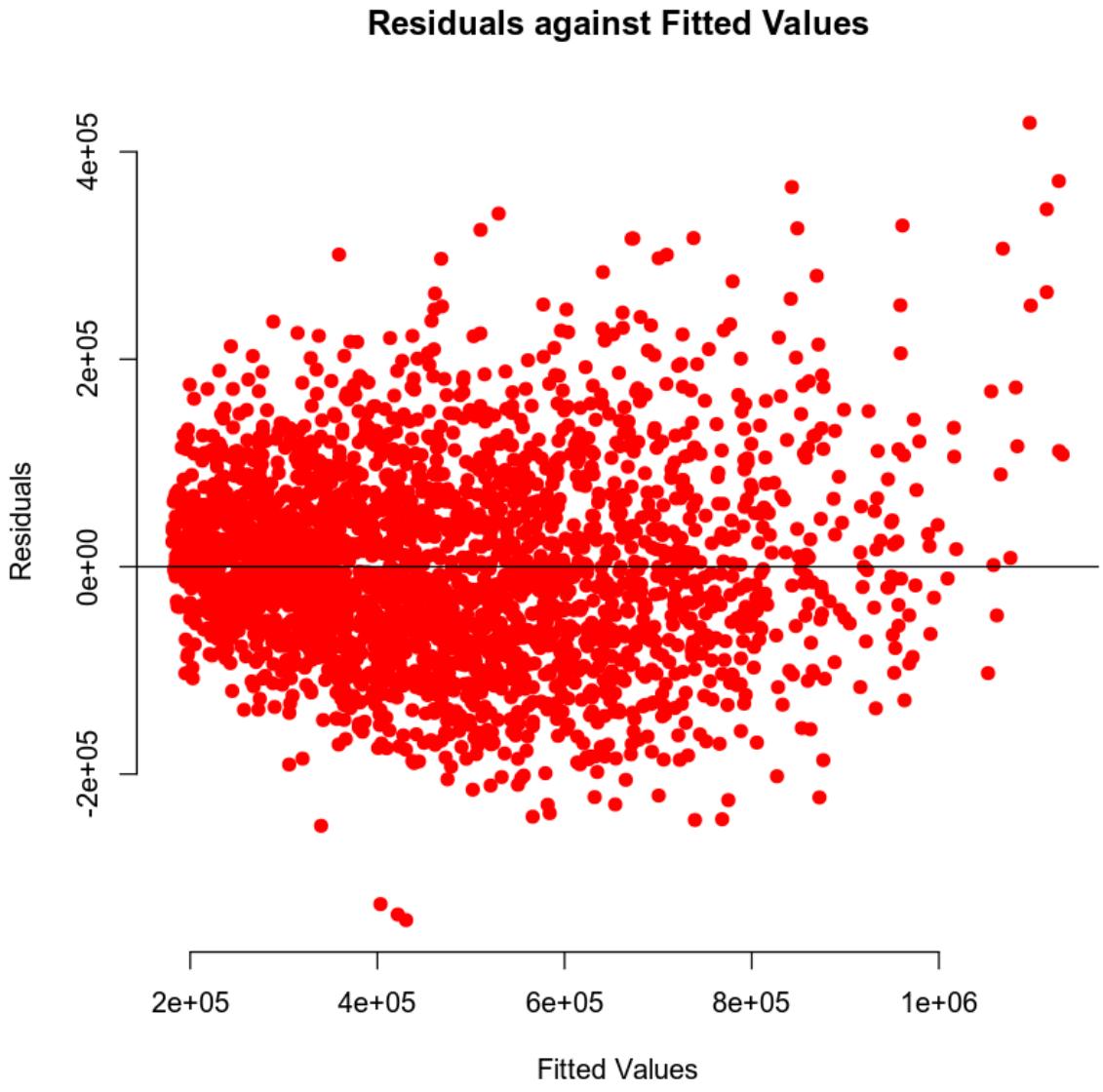
```
Call:
lm(formula = price ~ school_rating + crime + school_rating:crime +
    I(school_rating^2) + I(crime^2), data = housing)

Residuals:
    Min      1Q  Median      3Q     Max
-340729  -61055   -6288   56875  427915

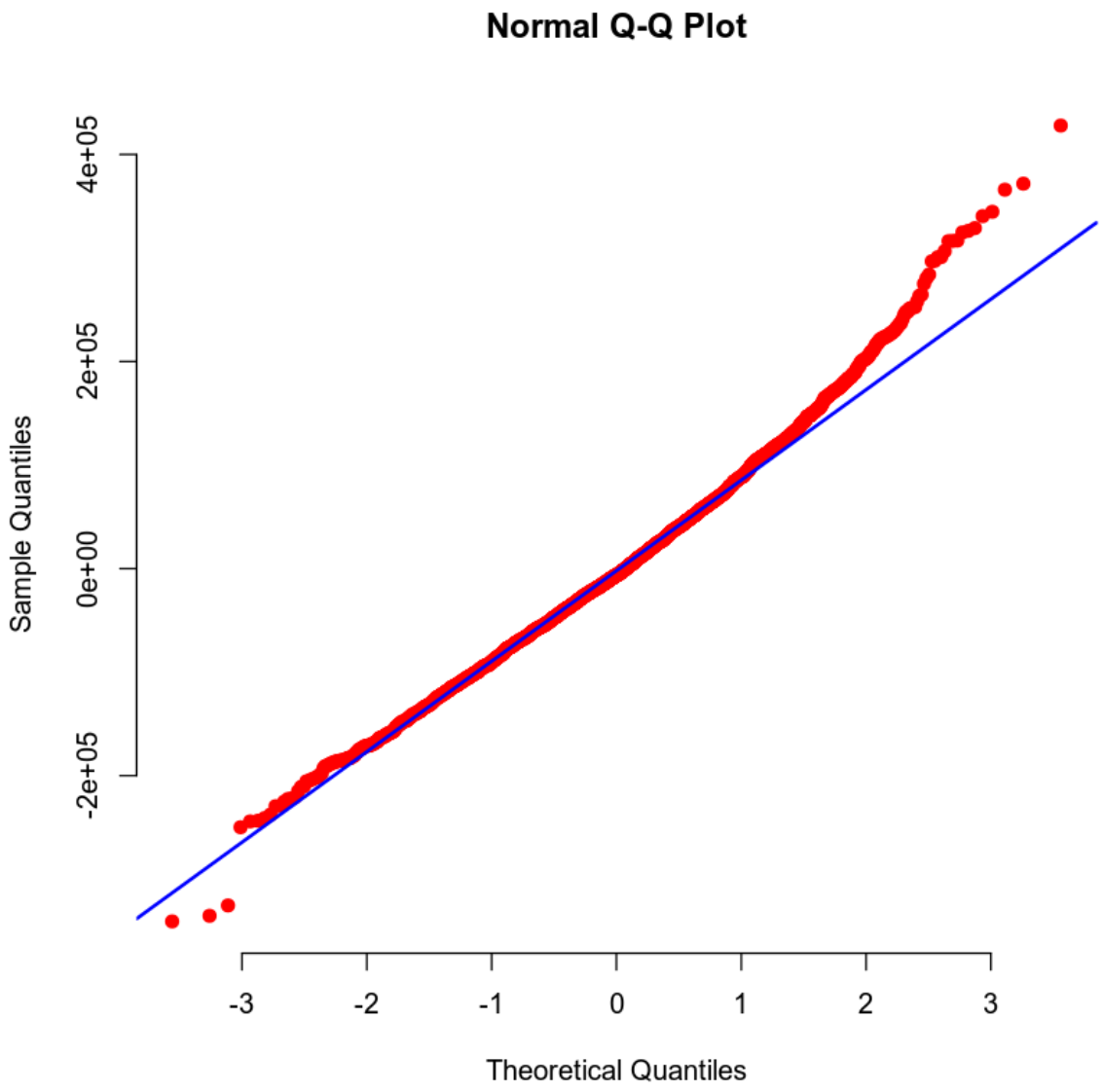
Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)          7.339e+05  1.032e+05   7.113 1.45e-12 ***
school_rating       -7.375e+04  2.083e+04  -3.541 0.000406 ***
crime               -3.155e+03  5.235e+02  -6.027 1.90e-09 ***
I(school_rating^2)   1.165e+04  1.109e+03  10.497  < 2e-16 ***
I(crime^2)           6.377e+00  7.265e-01   8.777  < 2e-16 ***
school_rating:crime -5.227e+01  4.853e+01  -1.077 0.281513
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 92690 on 2686 degrees of freedom
Multiple R-squared:  0.8088,    Adjusted R-squared:  0.8084
F-statistic:  2272 on 5 and 2686 DF,  p-value: < 2.2e-16
```

In [52]:
```
fitted_values <- fitted.values(model2)
residuals <- residuals(model2)
plot(fitted_values, residuals,
     main = "Residuals against Fitted Values",
     xlab = "Fitted Values", ylab = "Residuals",
     col="red",
     pch = 19, frame = FALSE)
abline(0,0)
```



In [53]:
```
qqnorm(residuals, pch = 19, col="red", frame = FALSE)
qqline(residuals, col = "blue", lwd = 2)
```



In [54]:
```
# Predicted price for a home in an area with average school rating of 9.80 and
# a crime rate of 81.02 per 100,000 individuals:
newdata <- data.frame(school_rating=9.80, crime=81.02)

print("prediction interval")
prediction_pred_int <- predict(model2, newdata, interval="predict", level=0.9
0)
round(prediction_pred_int, 4)

print("confidence interval")
prediction_conf_int <- predict(model2, newdata, interval="confidence", level=
0.90)
round(prediction_conf_int, 4)
```

```
[1] "prediction interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
874497	721606.2	1027388

```
[1] "confidence interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
874497	863681.4	885312.7

In [55]:
```
# Predicted price for a home in an area with average school rating of 4.28 and
# a crime rate of 215.50 per 100,000 individuals
newdata <- data.frame(school_rating=4.28, crime=215.50)

print("prediction interval")
prediction_pred_int <- predict(model2, newdata, interval="predict", level=0.9
0)
round(prediction_pred_int, 4)

print("confidence interval")
prediction_conf_int <- predict(model2, newdata, interval="confidence", level=
0.90)
round(prediction_conf_int, 4)
```

```
[1] "prediction interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
199706.7	46991.65	352421.7

```
[1] "confidence interval"
```

A matrix: 1 × 3 of type dbl

fit	lwr	upr
199706.7	191753.5	207659.9

Nested Models F-Test

Comparison of the reduced model (Model 3) and the complete second order model (Model #2 above).

In [56]:
```
# Create the first order regression model and print the statistics
model3 <- lm(price ~ school_rating + crime + school_rating:crime, data=housin
g)
summary(model3)
```

```
Call:
lm(formula = price ~ school_rating + crime + school_rating:crime,
    data = housing)

Residuals:
    Min      1Q  Median      3Q     Max
-336984  -63754   -4397   58894  440377

Coefficients:
                      Estimate Std. Error t value Pr(>|t|)
(Intercept)         -410233.37   25261.25  -16.24   <2e-16 ***
school_rating        155559.97    3133.06   49.65   <2e-16 ***
crime                  2230.07     129.70   17.20   <2e-16 ***
school_rating:crime    -564.85      17.86  -31.63   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 94870 on 2688 degrees of freedom
Multiple R-squared:  0.7995,    Adjusted R-squared:  0.7993
F-statistic:  3573 on 3 and 2688 DF,  p-value: < 2.2e-16
```

In [57]:
```
# Perform the F-test
anova(model2, model3)
```

A anova: 2 × 6

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
2686	2.307469e+13	NA	NA	NA	NA
2688	2.419501e+13	-2	-1.120319e+12	65.20513	2.22716e-28